Experiment Title: Installation and Configuration of Linux Operating System

Topic: Step-by-Step Installation of Ubuntu Linux via Bootable USB

1. Objective

The objectives of this experiment are:

- Understand the full procedure for installing a modern Linux operating system.
- Create a bootable installation media using an official Ubuntu ISO and Rufus.
- Configure disk partitioning (Root, Swap, EFI) and set up user account security.
- Verify successful operating system deployment using terminal-based commands.

2. Theoretical Background

Linux is a free, open-source Unix-like kernel released by Linus Torvalds in 1991. It provides multi-user security, high stability, and modular architecture, powering world servers, supercomputers, cloud infrastructure, and desktop systems.

2.1 Popular Linux Distributions

Distribution	Base System	Primary Focus / Use Case	Package Manager
Ubuntu	Debian	Beginner-friendly desktop, servers, and cloud	APT (`deb`)
Debian	Independent	Rock-solid stability and open-source standards	APT (`deb`)
Fedora	Independent	Cutting-edge packages and upstream Linux features	DNF / RPM
Kali Linux	Debian	Cybersecurity, penetration testing, and forensics	APT (`deb`)

2.2 Deployment Methods: Bare Metal vs Virtualization

Method	Simple Explanation
Bare Metal (Direct)	Installed directly onto the physical computer drive for maximum hardware performance.
Dual Boot	Installed alongside an existing OS (e.g., Windows), letting the user select OS at startup.
Virtual Machine (VM)	Runs inside virtualization software (e.g., VirtualBox), providing safe isolated testing.

3. Equipment / Software Requirements

Item	Details	Item	Details
Target Computer	PC / Laptop (64-bit x86 architecture)	**RAM Requirement**	Minimum 4 GB recommended
Storage Space	Minimum 25 GB free disk space	**Flash Drive**	8 GB+ USB Flash Drive
Target OS ISO	Ubuntu Desktop (64-bit ISO image)	**Flashing Tool**	Rufus / BalenaEtcher

4. Procedure / Steps

The installation was carried out sequentially through the following 10 steps:

1. Download the official Ubuntu Desktop ISO disk image.
2. Write the ISO to a USB flash drive using Rufus to create a bootable drive.
3. Boot from USB and select “Try or Install Ubuntu” from the GRUB menu.
4. Initialize live environment (Ubuntu splash screen and loading spinner).
5. Select keyboard layout and choose manual installation for custom partitioning.
6. Partition the drive: create EFI System Partition and Root (`/`) partition.
7. Configure user credentials: enter name, computer name, username, and password.
8. Review disk partitioning and configuration settings before confirming.
9. Execute system installation, package copying, and driver setup.
10. Complete installation, remove the USB drive, and reboot into Ubuntu.

4.1 Step 1: Downloading Ubuntu ISO Image

Visit the official Ubuntu website (`https://ubuntu.com/download/desktop`) and download the latest Ubuntu Desktop 64-bit ISO file. This image contains the complete Linux operating system installer and live environment.

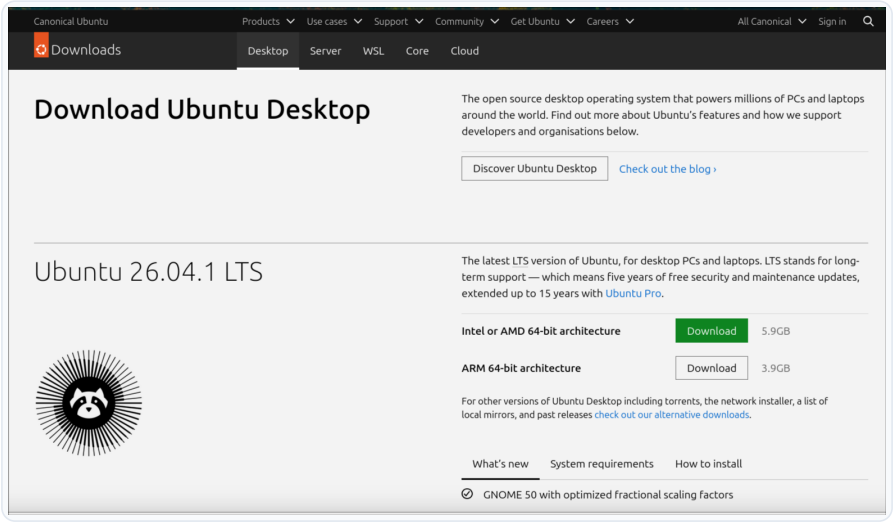


Figure 1: Downloading Ubuntu Desktop ISO from Official Website

4.2 Step 2: Creating a Bootable USB Flash Drive

Insert a USB flash drive (minimum 8 GB) into the host computer. Open Rufus, select the downloaded Ubuntu ISO file, choose **GPT** partition scheme for UEFI systems, and click **START** to write the ISO image to the flash drive.

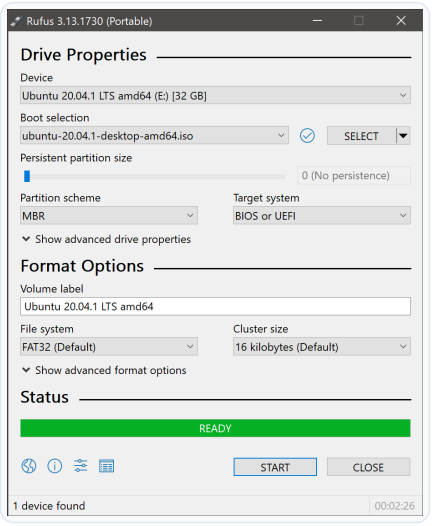


Figure 2: Writing ISO Image to USB Flash Drive Using Rufus

4.3 Step 3: Selecting "Try or Install Ubuntu"

Insert the bootable USB and restart the computer. Press the Boot Menu key (e.g., **F12**, **F9**, or **Esc**) and select the USB drive. The GNU GRUB bootloader appears on screen. Select **Try or Install Ubuntu** and press **Enter**.



Figure 3: GRUB Boot Menu – Selecting Try or Install Ubuntu

4.4 Step 4: Booting into the Live Environment

After selecting the installation option, the Ubuntu boot splash screen appears with the official logo and spinner while the operating system initializes the live environment into RAM.

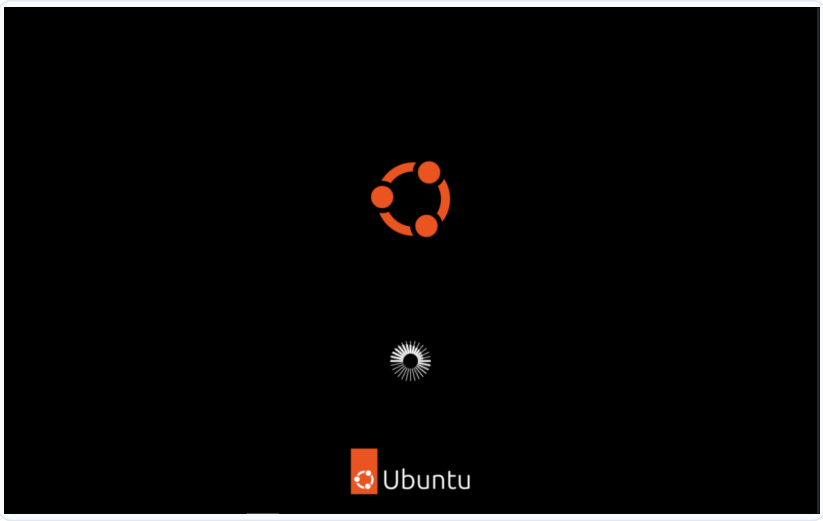


Figure 4: Ubuntu Boot Splash Screen Initializing from USB

4.5 Step 5: Selecting Manual Installation Type

Choose system language and keyboard layout. When prompted for installation type, choose **Manual installation** (or **Something else**) to maintain full control over disk partitions and filesystem layout.

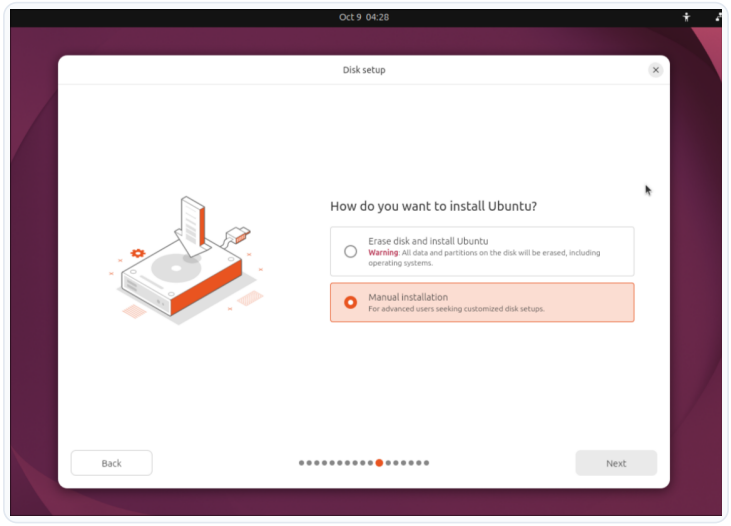


Figure 5: Selecting Manual Installation for Custom Partitioning

4.6 Step 6: Disk Partitioning

Allocate drive space for the operating system:

- **EFI System Partition:** Size 500 MB (used by UEFI firmware to load the GRUB bootloader).
- **Root Partition (/):** Remaining space formatted as **Ext4** journaling filesystem.

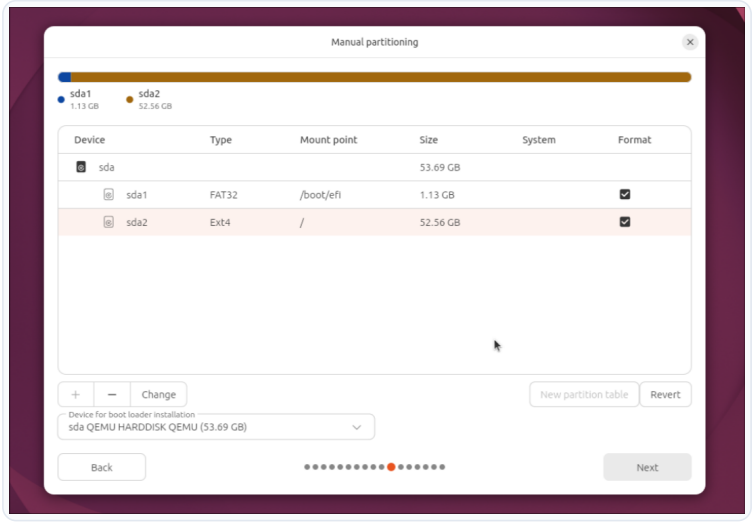


Figure 6: Configuring EFI and Root Disk Partitions

4.7 Step 7: Creating User Account and Password

Set up user identification and system security:

- Enter your full name and computer hostname.
- Choose a username and a strong security password.
- Select **Require my password to log in** for administrative privilege protection.

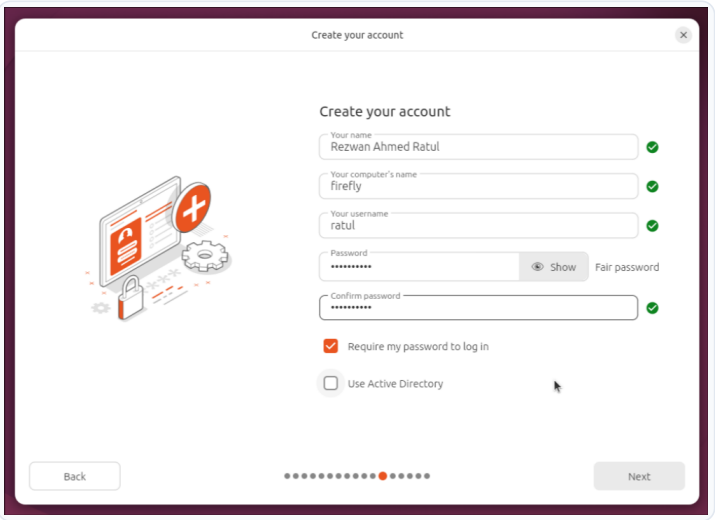


Figure 7: Setting Up Username, Computer Hostname, and Password

4.8 Step 8: Reviewing Installation Configuration

Before writing changes to disk, the installer presents a summary of selected partitions, mount points, and timezone settings. Review the list and click **Install** to confirm changes and begin file writing.

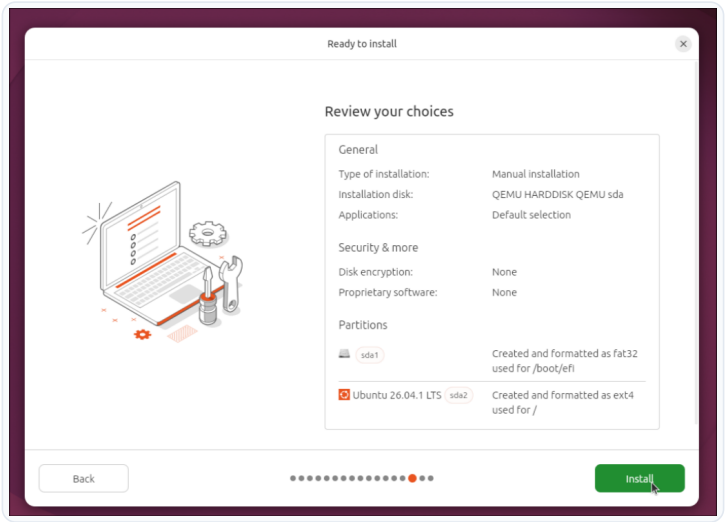


Figure 8: Reviewing Partition Summary and Confirming Installation

4.9 Step 9: System Installation Process

The installer formats partitions, copies core Linux system files, configures hardware drivers, and sets up localized packages. A progress bar displays real-time setup status.

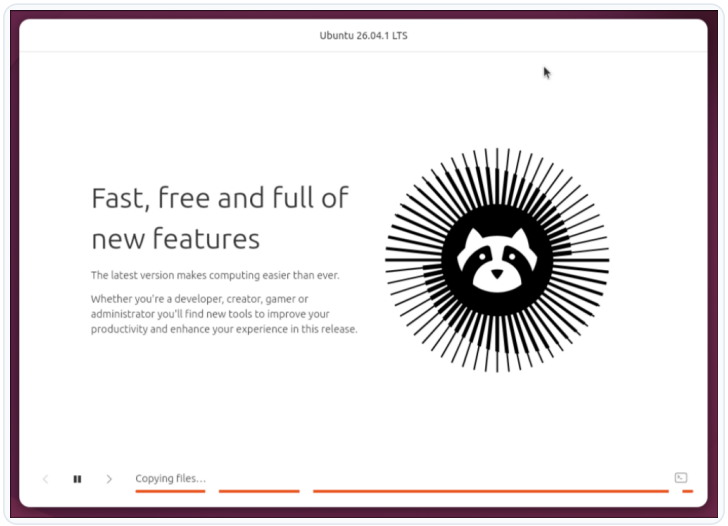


Figure 9: Automated System Installation and File Extraction

4.10 Step 10: Completing Installation and System Reboot

Once file copying completes, an **"Installation is complete"** notification appears. Click **Restart Now**. When prompted by the screen, unplug the USB flash drive and press **Enter** to restart.

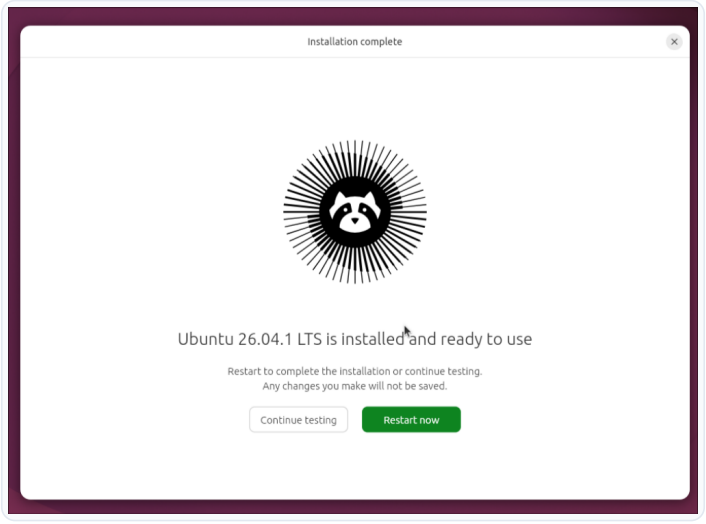


Figure 10: Installation Finished Notification and Reboot Prompt

5. Result / Output Demonstration

The computer restarts directly into the newly installed Ubuntu Linux system.

5.1 Desktop Interface Demonstration

The machine boots cleanly into the GNOME Desktop Environment, showing the top panel, system dock, application drawer, and desktop workspace.



Figure 11: Ubuntu Desktop Environment Loaded After First Boot

5.2 Terminal-Based System Verification

To confirm proper operating system installation, kernel functionality, and hardware allocation, we opened the Linux terminal and executed standard verification commands:

```
ratul@firefly: ~ — bash

ratul@firefly:~$ uname -a
Linux firefly 6.8.0-45-generic #45-Ubuntu SMP PREEMPT_DYNAMIC Wed Sep 11 15:25:05 UTC 2024 x86_64 GNU/Linux
ratul@firefly:~$ lsb_release -a
Distributor ID: Ubuntu Description: Ubuntu 24.04.1 LTS Release: 24.04 Codename: noble
ratul@firefly:~$ whoami
ratul
ratul@firefly:~$ df -h /
Filesystem Size Used Avail Use% Mounted on /dev/nvme0n1p2 59G 8.2G 48G 15% /
ratul@firefly:~$ _
```

Explanation of Verification Outputs:

- `uname -a`: Confirmed Linux kernel version `6.8.0-45-generic` on 64-bit architecture.
- `lsb_release -a`: Verified official distribution name (`Ubuntu 24.04.1 LTS noble`).
- `whoami`: Confirmed active session under newly created user `ratul`.
- `df -h /`: Confirmed root filesystem mounted on `/dev/nvme0n1p2` with 48 GB available space.

6. Conclusion

Key takeaways from this lab:

1. Successfully created a bootable USB drive using Rufus and booted into Ubuntu via UEFI.
2. Configured manual disk partitioning with dedicated EFI and Ext4 Root filesystems.
3. Created a secure user account and verified system health through terminal commands.